



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Linux Command Line For Digital Forensics

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Course Title: Basic Computer Skills for Digital Forensics

Course Code: SBT-DF201

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1. INTRODUCTION

Digital forensics requires direct, non-destructive interaction with file systems, metadata, and logs to ensure evidence integrity is maintained. This lab establishes core command-line techniques such as workspace initialization, directory traversal, text pattern matching, safe networking diagnostics, script automation, and archiving to equip analysts for live response and artifact recovery.

2. Lab Environment & Setup

- **Operating System:** Kali Linux Rolling (Version 2026.3)
- **Kernel & Architecture:** Linux 7.0.12+kali-amd64 on x86_64
- **User Context:** garba (UID 1000) with standard sudo privileges
- **Core Utilities Used:** tree, nmap, wget, zip, unzip, tar, net-tools, terminator, nano, grep, stat, ss, ip, find
- **Package Staging:** Missing utilities were resolved via repository synchronization and apt install.

3. Tasks Performed & Command Execution

Prepare Your Evidence Workspace

Create a structured workspace to hold all commands, outputs, screenshots, and final evidence artifacts.

Commands Executed:

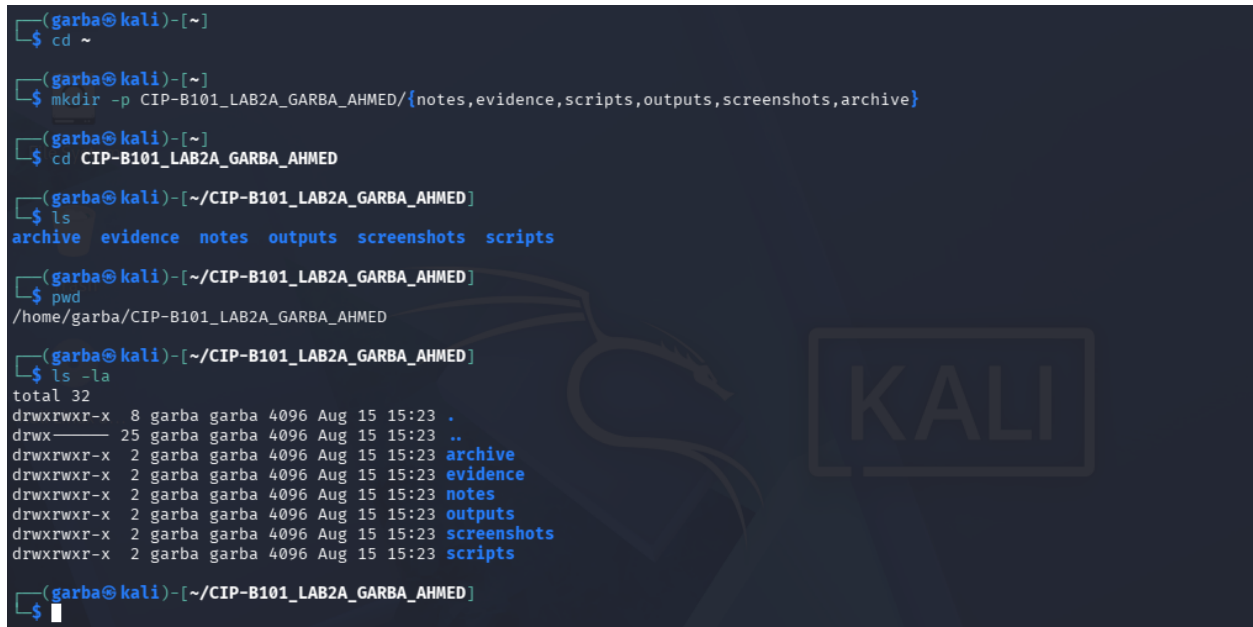
```
cd ~
```

```
mkdir -p CIP-B101_Lab2A_Garba_Ahmed/{notes,evidence,scripts,outputs,screenshots,archive}
```

```
cd CIP-B101_Lab2A_Garba_Ahmed
```

```
pwd
```

```
ls -la
```

A terminal window with a dark background and a 'KALI' logo watermark. The terminal shows the following commands and their outputs:

```
(garba@kali)-[~]  
$ cd ~  
  
(garba@kali)-[~]  
$ mkdir -p CIP-B101_LAB2A_GARBA_AHMED/{notes,evidence,scripts,outputs,screenshots,archive}  
  
(garba@kali)-[~]  
$ cd CIP-B101_LAB2A_GARBA_AHMED  
  
(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]  
$ ls  
archive  evidence  notes  outputs  screenshots  scripts  
  
(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]  
$ pwd  
/home/garba/CIP-B101_LAB2A_GARBA_AHMED  
  
(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]  
$ ls -la  
total 32  
drwxrwxr-x 8 garba garba 4096 Aug 15 15:23 .  
drwx----- 25 garba garba 4096 Aug 15 15:23 ..  
drwxrwxr-x 2 garba garba 4096 Aug 15 15:23 archive  
drwxrwxr-x 2 garba garba 4096 Aug 15 15:23 evidence  
drwxrwxr-x 2 garba garba 4096 Aug 15 15:23 notes  
drwxrwxr-x 2 garba garba 4096 Aug 15 15:23 outputs  
drwxrwxr-x 2 garba garba 4096 Aug 15 15:23 screenshots  
drwxrwxr-x 2 garba garba 4096 Aug 15 15:23 scripts  
  
(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]  
$
```

Task 2: Inspect Linux Filesystem and OS Information

Observe the Linux filesystem hierarchy, operating system release details, and basic execution privileges.

Commands Executed:

```
clear
```

```
pwd
```

```
ls /
```

```
ls -la /
```

```
cat /etc/os-release
```

```
uname -a
```

```
whoami
```

```
id
```

- **Evidence Screenshots:**

```
(garba@kali) - [~/CIP-B101_LAB2A_GARBA_AHMED]
$ pwd
/home/garba/CIP-B101_LAB2A_GARBA_AHMED

(garba@kali) - [~/CIP-B101_LAB2A_GARBA_AHMED]
$ ls /
bin  dev  home  initrd.img.old  lib32  lost+found  mnt  proc  run  srv  tmp  var  vmlinuz.old
boot  etc  initrd.img  lib  lib64  media  opt  root  sbin  sys  usr  vmlinuz

(garba@kali) - [~/CIP-B101_LAB2A_GARBA_AHMED]
$ cat /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
NAME="Kali GNU/Linux"
VERSION_ID="2026.3"
VERSION="2026.3"
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"

(garba@kali) - [~/CIP-B101_LAB2A_GARBA_AHMED]
$ uname -a
Linux kali 7.0.12-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 7.0.12-2kali1 (2026-06-18) x86_64 GNU/Linux

(garba@kali) - [~/CIP-B101_LAB2A_GARBA_AHMED]
$ whoami
garba

(garba@kali) - [~/CIP-B101_LAB2A_GARBA_AHMED]
$ id
uid=1000(garba) gid=1000(garba) groups=1000(garba),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(netdev),103(scanner),116(bluetooth),121(lpadmin),124(wireshark),130(kaboxer)
```

Task 3: Understand Inodes and Disk Information

Create a test evidence file and inspect its inode, system metadata, and storage constraints.

Commands Executed:

```
echo "This is Lab 2A evidence file created by Garba Ahmed" > evidence/lab2a_evidence.txt
```

```
ls -li evidence/lab2a_evidence.txt
```

stat evidence/lab2a_evidence.txt

df -h

df -i

```
(garba@kali) [~/CIP-B101_LAB2A_GARBA_AHMED]
$ echo "This is Lab 2A evidence file created by Garba Ahmed" > evidence/lab2a_evidence.txt

(garba@kali) [~/CIP-B101_LAB2A_GARBA_AHMED]
$ ls -li evidence/lab2a_evidence.txt
787053 -rw-rw-r-- 1 garba garba 52 Aug 15 15:45 evidence/lab2a_evidence.txt

(garba@kali) [~/CIP-B101_LAB2A_GARBA_AHMED]
$ stat evidence/lab2a_evidence.txt
File: evidence/lab2a_evidence.txt
Size: 52      Blocks: 8      IO Block: 4096   regular file
Device: 8,1   Inode: 787053   Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   garba)   Gid: ( 1000/   garba)
Access: 2026-08-15 15:45:45.627031523 +0100
Modify: 2026-08-15 15:45:45.627031523 +0100
Change: 2026-08-15 15:45:45.627031523 +0100
Birth: 2026-08-15 15:45:45.627031523 +0100

(garba@kali) [~/CIP-B101_LAB2A_GARBA_AHMED]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            861M   0 861M   0% /dev
tmpfs           194M  1.2M  193M   1% /run
/dev/sda1       43G   21G   20G  52% /
tmpfs           967M  4.0K  967M   1% /dev/shm
none            1.0M   0 1.0M   0% /run/credentials/systemd-journald.service
tmpfs           967M  60K  967M   1% /tmp
none            1.0M   0 1.0M   0% /run/credentials/getty@tty1.service
tmpfs           194M  108K  194M   1% /run/user/1000

(garba@kali) [~/CIP-B101_LAB2A_GARBA_AHMED]
$ df -i
Filesystem      Inodes  IUsed  IFree IUse% Mounted on
udev            220306   408  219898    1% /dev
tmpfs           247339   767  246572    1% /run
/dev/sda1       2818048 607774 2210274   22% /
tmpfs           247339    2  247337    1% /dev/shm
none            1024    1   1023    1% /run/credentials/systemd-journald.service
tmpfs          1048576   22 1048554    1% /tmp
none            1024    1   1023    1% /run/credentials/getty@tty1.service
tmpfs           49467   179   49288    1% /run/user/1000
```

Task 4: Absolute and Relative Paths

Demonstrate fluency with absolute and relative paths by navigating directories via the command line.

Commands Executed:

```
cd ~/CIP-B101_Lab2A_Garba_Ahmed/evidence
```

```
pwd
```

```
cd ..
```

```
pwd
```

```
cd ./notes
```

```
pwd
```

```
cd ../evidence
```

```
pwd
```

readlink -f lab2a_evidence.txt

```
(garba@kali)-[~]
└─$ cd ~/CIP-B101_LAB2A_GARBA_AHMED/notes
(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED/notes]
└─$ pwd
/home/garba/CIP-B101_LAB2A_GARBA_AHMED/notes
(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED/notes]
└─$ ls
(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED/notes]
└─$ cd ../evidence
(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED/evidence]
└─$ pwd
/home/garba/CIP-B101_LAB2A_GARBA_AHMED/evidence
```

Task 5: Create, View, Copy, Rename and Remove Files

Execute basic file management workflows standard for organizing working copies and forensic extractions.

Commands Executed:

```
cd ~/CIP-B101_Lab2A_Garba_Ahmed
```

```
echo "Case Note 1: Initial Linux command line practice" > notes/case_note1.txt
```

```
echo "Case Note 2: Evidence handling requires careful documentation" > notes/case_note2.txt
```

```
cat notes/case_note1.txt
```

```
cat notes/case_note2.txt
```

```
cp notes/case_note1.txt evidence/case_note1_copy.txt
```

```
cp -r notes evidence/notes_backup
```

```
mv evidence/case_note1_copy.txt evidence/case_note1_renamed.txt
```

```
ls -la evidence
```

```
echo "temporary data" > evidence/temp_delete_me.txt
```

```
ls -la evidence/temp_delete_me.txt
```

```
rm evidence/temp_delete_me.txt
```

```
ls -la evidence
```

```

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED/evidence]
$ cd ~/CIP-B101_LAB2A_GARBA_AHMED

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ echo "Case Note 1: Initial Linux command line practice" > notes/case_note1.txt

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ echo "Case Note 2: Evidence handling requires careful documentation" > notes/case_note2.txt

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ cat notes/case_note1.txt
Case Note 1: Initial Linux command line practice

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ cat notes/case_note2.txt
Case Note 2: Evidence handling requires careful documentation

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ cp notes/case_note1.txt evidence/case_note1_copy.txt

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ cp notes/case_note2.txt evidence/case_note2_copy.txt

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ cp -r notes evidence/notes_backup

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ mv evidence/case_note1_copy.txt evidence/case_note1_renamed.txt
v: command not found

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ ls -la evidence
total 20
drwxrwxr-x 3 garba garba 4096 Aug 15 16:02 .
drwxrwxr-x 8 garba garba 4096 Aug 15 15:23 ..
-rw-rw-r-- 1 garba garba 49 Aug 15 16:01 case_note1_copy.txt
-rw-rw-r-- 1 garba garba 52 Aug 15 15:45 lab2a_evidence.txt
drwxrwxr-x 2 garba garba 4096 Aug 15 16:02 notes_backup

```

Task 6: Search Evidence Content with grep

Parse text-based logs using structured heredocs and extract investigative indicators using regular expressions.

Commands Executed:

```
cat > evidence/activity_log.txt <<'EOF'
```

```
2026-06-19 09:10 user=aminu action=login status=success ip=192.168.56.10
```

```
2026-06-19 09:25 user=guest action=file_copy status=success file=report.docx
```

```
2026-06-19 10:05 user=admin action=login status=failed ip=192.168.56.15
```

```
2026-06-19 10:20 user=aminu action=delete status=success file=temp.txt
```

```
2026-06-19 10:40 user=admin action=password_change status=success
```

```
EOF
```

```
cat evidence/activity_log.txt
```

```
grep "login" evidence/activity_log.txt
```

```
grep -n "failed" evidence/activity_log.txt
```

```
grep -Ei "delete|password" evidence/activity_log.txt
```

```
grep -Ei "192\.168\.56\.[0-9]+" evidence/activity_log.txt
```

```

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ cat > evidence/activity_log.txt <<'EOF'
heredoc> 2026-06-19 09:10 user=aminu action=login status=success ip=192.168.56.10
heredoc> 2026-06-19 09:25 user=guest action=file_copy status=success file=report.docx
heredoc> 2026-06-19 10:05 user=admin action=login status=failed ip=192.168.56.15
heredoc> 2026-06-19 10:20 user=aminu action=delete status=success file=temp.txt
heredoc> 2026-06-19 10:40 user=admin action=password_change status=success
heredoc> EOF

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ cat evidence/activity_log.txt
2026-06-19 09:10 user=aminu action=login status=success ip=192.168.56.10
2026-06-19 09:25 user=guest action=file_copy status=success file=report.docx
2026-06-19 10:05 user=admin action=login status=failed ip=192.168.56.15
2026-06-19 10:20 user=aminu action=delete status=success file=temp.txt
2026-06-19 10:40 user=admin action=password_change status=success

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ grep "login" evidence/activity_log.txt
2026-06-19 09:10 user=aminu action=login status=success ip=192.168.56.10
2026-06-19 10:05 user=admin action=login status=failed ip=192.168.56.15

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ grep -Ei "delete|password" evidence/activity_log.txt
2026-06-19 10:20 user=aminu action=delete status=success file=temp.txt
2026-06-19 10:40 user=admin action=password_change status=success

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ grep -Ei "192\.168\.56\.[0-9]+" evidence/activity_log.txt
2026-06-19 09:10 user=aminu action=login status=success ip=192.168.56.10
2026-06-19 10:05 user=admin action=login status=failed ip=192.168.56.15

```

Task 7: Search File Names and Display Directory Tree

Map out workspace layouts using tree and locate files dynamically using name and pattern searches with find.

Commands Executed:

```
cd ~/CIP-B101_Lab2A_Garba_Ahmed
```

```
tree
```

```
find . -type f -name "*.txt"
```

```
find . -type f -iname "*log*"
```

```
find . -type d -name "evidence"
```



```
(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ tree
.
├── archive
├── evidence
│   ├── activity_log.txt
│   ├── case_note1_copy.txt
│   ├── lab2a_evidence.txt
│   └── notes_backup
│       ├── case_note1.txt
│       └── case_note2.txt
├── notes
│   ├── case_note1.txt
│   └── case_note2.txt
├── outputs
├── screenshots
└── scripts

8 directories, 7 files

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ find . -type f -name "*.txt"
./evidence/case_note1_copy.txt
./evidence/activity_log.txt
./evidence/lab2a_evidence.txt
./evidence/notes_backup/case_note1.txt
./evidence/notes_backup/case_note2.txt
./notes/case_note1.txt
./notes/case_note2.txt

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ find . -type f -iname "*log*"
./evidence/activity_log.txt

(garba@kali)-[~/CIP-B101_LAB2A_GARBA_AHMED]
$ find . -type d -name "evidence"
./evidence
```

Task 8: Check Network Information Safely

Inspect local network configurations, routing tables, loopback availability, open ports, and local socket listeners safely.

Commands Executed:

ip addr

ip route

ping -c 4 127.0.0.1

ping -c 4 localhost

ss -lnt

nmap localhost -p 22,80,443 2>/dev/null || echo "nmap unavailable or no services detected"

```
(garba@kali)~/CIP-B101_LAB2A_GARBA_AHMED
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:18:64:f5 brd ff:ff:ff:ff:ff:ff
    inet 192.168.88.131/24 brd 192.168.88.255 scope global dynamic noprefixroute eth0
        valid_lft 1604sec preferred_lft 1604sec
    inet6 fe80::20c:29ff:fe18:64f5/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether f6:fa:73:39:63:09 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

(garba@kali)~/CIP-B101_LAB2A_GARBA_AHMED
$ ip route
default via 192.168.88.2 dev eth0 proto dhcp src 192.168.88.131 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.88.0/24 dev eth0 proto kernel scope link src 192.168.88.131 metric 100

(garba@kali)~/CIP-B101_LAB2A_GARBA_AHMED
$ ping -c 4 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data:
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=6.45 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=14.9 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.073 ms
^C
--- 127.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2005ms
rtt min/avg/max/mdev = 0.073/7.128/14.860/6.055 ms

(garba@kali)~/CIP-B101_LAB2A_GARBA_AHMED
$ ping -c 4 localhost
PING localhost (::1) 56 data bytes
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.073 ms
^C
--- 127.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2005ms
rtt min/avg/max/mdev = 0.073/7.128/14.860/6.055 ms

(garba@kali)~/CIP-B101_LAB2A_GARBA_AHMED
$ ping -c 4 localhost
PING localhost (::1) 56 data bytes
64 bytes from localhost (::1): icmp_seq=1 ttl=64 time=1.75 ms
64 bytes from localhost (::1): icmp_seq=2 ttl=64 time=13.9 ms
64 bytes from localhost (::1): icmp_seq=3 ttl=64 time=0.055 ms
64 bytes from localhost (::1): icmp_seq=4 ttl=64 time=0.079 ms
--- localhost ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3028ms
rtt min/avg/max/mdev = 0.055/3.949/13.913/5.793 ms

(garba@kali)~/CIP-B101_LAB2A_GARBA_AHMED
$ ss -ltn
State          Recv-Q          Send-Q           Local Address:Port    Peer Address:Port
LISTEN         0                4096             127.0.0.1:35399       0.0.0.0:*

(garba@kali)~/CIP-B101_LAB2A_GARBA_AHMED
$ nmap localhost -p 22,80,443 2>/dev/null || echo "nmap unavailable or no services detected"
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-16 09:55 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0018s latency).
Other addresses for localhost (not scanned): ::1

PORT      STATE SERVICE
22/tcp    closed ssh
80/tcp    closed http
443/tcp   closed https

Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
```

Task 9: Create and Execute a Simple Forensic Script

Build an automated shell script to record system states, disk metrics, and folder trees into an output report.

Script Configuration (scripts/lab2a_sys_summary.sh):

```
#!/bin/sh

# CIP-B101 Lab 2A system summary script

# Student: Garba Ahmed

OUT="$HOME/CIP-B101_Lab2A_Garba_Ahmed/outputs/system_summary.txt"

{
    echo "==== CIP-B101 Lab 2A System Summary ====="
    echo "Date/Time: $(date)"
    echo "User: $(whoami)"
    echo "Hostname: $(hostname)"
    echo "Kernel: $(uname -a)"
    echo "Current Folder: $(pwd)"
    echo "==== Disk Space ====="
    df -h
    echo "==== Inode Usage ====="
    df -i
    echo "==== Evidence Workspace Tree ====="
    tree "$HOME/CIP-B101_Lab2A_Garba_Ahmed"
} > "$OUT"

echo "Summary saved to $OUT"
```

- **Commands Executed:**

Bash

```
cd ~/CIP-B101_Lab2A_Garba_Ahmed/scripts
```

```
chmod +x lab2a_sys_summary.sh
```

```
ls -l lab2a_sys_summary.sh
```

```
./lab2a_sys_summary.sh
```

```
cat ../outputs/system_summary.txt | head -n 30
```

```
GNU nano 9.1 lab2a_sys_summary.sh *
#!/bin/sh
# CIP-B101 Lab 2A system summary script
# Student: FIRSTNAME LASTNAME

OUT="$HOME/CIP-B101_Lab2A_FIRSTNAME_LASTNAME/outputs/system_summary.txt"
{
  echo "===== CIP-B101 Lab 2A System Summary ====="
  echo "Date/Time: $(date)"
  echo "User: $(whoami)"
  echo "Hostname: $(hostname)"
  echo "Kernel: $(uname -a)"
  echo "Current Folder: $(pwd)"
  echo "===== Disk Space ====="
  df -h
  echo "===== Inode Usage ====="
  df -i
  echo "===== Evidence Workspace Tree ====="
  tree "$HOME/CIP-B101_Lab2A_FIRSTNAME_LASTNAME"
} > "$OUT"

echo "Summary saved to $OUT"
chmod +x lab2a_sys_summary.sh
ls -l lab2a_sys_summary.sh
./lab2a_sys_summary.sh
cat ../outputs/system_summary.txt | head -n 30
```

Task 10: Make the Script Callable from Another Folder Using PATH

Temporarily modify the PATH environment variable to execute custom scripts globally from any directory.

Commands Executed:

```
export PATH=$HOME/CIP-B101_Lab2A_Garba_Ahmed/scripts:$PATH
```

```
cd /tmp
```

```
lab2a_sys_summary.sh
```

```
cat ~/CIP-B101_Lab2A_Garba_Ahmed/outputs/system_summary.txt | head -n 15
```

Task 11: Compress the Evidence Workspace

Bundle and compress the finalized investigative workspace using standard archiving tools for transport.

Commands Executed:

```
cd ~
```

```
zip -r CIP-B101_Lab2A_Garba_Ahmed.zip CIP-B101_Lab2A_Garba_Ahmed
```

```
tar -czvf CIP-B101_Lab2A_Garba_Ahmed.tar.gz CIP-B101_Lab2A_Garba_Ahmed
```

```
ls -lh CIP-B101_Lab2A_Garba_Ahmed.zip CIP-B101_Lab2A_Garba_Ahmed.tar.gz
```

```
(garba@kali)-[~]
$ tar -czvf CIP-B101_LAB2A_GARBA_AHMED.tar.gz CIP-B101_LAB2A_GARBA_AHMED
CIP-B101_LAB2A_GARBA_AHMED/
CIP-B101_LAB2A_GARBA_AHMED/screenshots/
CIP-B101_LAB2A_GARBA_AHMED/scripts/
CIP-B101_LAB2A_GARBA_AHMED/evidence/
CIP-B101_LAB2A_GARBA_AHMED/evidence/case_note1_copy.txt
CIP-B101_LAB2A_GARBA_AHMED/evidence/activity_log.txt
CIP-B101_LAB2A_GARBA_AHMED/evidence/lab2a_evidence.txt
CIP-B101_LAB2A_GARBA_AHMED/evidence/notes_backup/
CIP-B101_LAB2A_GARBA_AHMED/evidence/notes_backup/case_note1.txt
CIP-B101_LAB2A_GARBA_AHMED/evidence/notes_backup/case_note2.txt
CIP-B101_LAB2A_GARBA_AHMED/notes/
CIP-B101_LAB2A_GARBA_AHMED/notes/case_note1.txt
CIP-B101_LAB2A_GARBA_AHMED/notes/case_note2.txt
CIP-B101_LAB2A_GARBA_AHMED/archive/
CIP-B101_LAB2A_GARBA_AHMED/lab2a_sys_summary.sh
CIP-B101_LAB2A_GARBA_AHMED/outputs/

(garba@kali)-[~]
$ ls -lh CIP-B101_LAB2A_GARBA_AHMED.zip CIP-B101_LAB2A_GARBA_AHMED.tar.gz
-rw-rw-r-- 1 garba garba 1.1K Aug 16 10:07 CIP-B101_LAB2A_GARBA_AHMED.tar.gz
-rw-rw-r-- 1 garba garba 4.3K Aug 16 10:05 CIP-B101_LAB2A_GARBA_AHMED.zip
```

```
(garba@kali)-[~]
$ cd ~

(garba@kali)-[~]
$ zip -r CIP-B101_Lab2A_GARBA_AHMED.zip CIP-B101_Lab2A_GARBA_AHMED
zip warning: name not matched: CIP-B101_Lab2A_GARBA_AHMED

zip error: Nothing to do! (try: zip -r CIP-B101_Lab2A_GARBA_AHMED.zip . -i CIP-B101_Lab2A_GARBA_AHMED)

(garba@kali)-[~]
$ zip -r CIP-B101_LAB2A_GARBA_AHMED.zip CIP-B101_LAB2A_GARBA_AHMED
adding: CIP-B101_LAB2A_GARBA_AHMED/ (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/screenshots/ (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/scripts/ (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/evidence/ (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/evidence/case_note1_copy.txt (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/evidence/activity_log.txt (deflated 55%)
adding: CIP-B101_LAB2A_GARBA_AHMED/evidence/lab2a_evidence.txt (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/evidence/notes_backup/ (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/evidence/notes_backup/case_note1.txt (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/evidence/notes_backup/case_note2.txt (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/notes/ (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/notes/case_note1.txt (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/notes/case_note2.txt (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/archive/ (stored 0%)
adding: CIP-B101_LAB2A_GARBA_AHMED/lab2a_sys_summary.sh (deflated 51%)
adding: CIP-B101_LAB2A_GARBA_AHMED/outputs/ (stored 0%)

(garba@kali)-[~]
$
```

```
(garba@kali)-[~]
$ tar -czvf CIP-B101_LAB2A_GARBA_AHMED.tar.gz CIP-B101_LAB2A_GARBA_AHMED
CIP-B101_LAB2A_GARBA_AHMED/
CIP-B101_LAB2A_GARBA_AHMED/screenshots/
CIP-B101_LAB2A_GARBA_AHMED/scripts/
CIP-B101_LAB2A_GARBA_AHMED/evidence/
CIP-B101_LAB2A_GARBA_AHMED/evidence/case_note1_copy.txt
CIP-B101_LAB2A_GARBA_AHMED/evidence/activity_log.txt
CIP-B101_LAB2A_GARBA_AHMED/evidence/lab2a_evidence.txt
CIP-B101_LAB2A_GARBA_AHMED/evidence/notes_backup/
CIP-B101_LAB2A_GARBA_AHMED/evidence/notes_backup/case_note1.txt
CIP-B101_LAB2A_GARBA_AHMED/evidence/notes_backup/case_note2.txt
CIP-B101_LAB2A_GARBA_AHMED/notes/
CIP-B101_LAB2A_GARBA_AHMED/notes/case_note1.txt
CIP-B101_LAB2A_GARBA_AHMED/notes/case_note2.txt
CIP-B101_LAB2A_GARBA_AHMED/archive/
CIP-B101_LAB2A_GARBA_AHMED/lab2a_sys_summary.sh
CIP-B101_LAB2A_GARBA_AHMED/outputs/

(garba@kali)-[~]
$ ls -lh CIP-B101_LAB2A_GARBA_AHMED.zip CIP-B101_LAB2A_GARBA_AHMED.tar.gz
-rw-rw-r-- 1 garba garba 1.1K Aug 16 10:07 CIP-B101_LAB2A_GARBA_AHMED.tar.gz
-rw-rw-r-- 1 garba garba 4.3K Aug 16 10:05 CIP-B101_LAB2A_GARBA_AHMED.zip
```

4. Forensic Interpretation & Technical Insights

Inodes: An inode is a kernel data structure that stores file metadata such as permissions, timestamps, ownership, and disk block pointers while explicitly excluding the file name. In digital forensics, inodes are critical because they persist independently of the file name, enabling analysts to track, analyze, and recover files even after they have been renamed or unlinked.

- **The PATH Variable:** The PATH environment variable defines an ordered list of directory paths that the shell scans to locate executable binaries when a command is invoked. Configuring custom script directories allows analysts to execute specialized tools globally without specifying absolute paths.
- **Grep Pattern Filtering:** Regular expression and pattern matching via grep allow investigators to instantly isolate indicators of compromise (IoCs), error codes, or specific transactional log entries out of large datasets.
- **Localhost Testing:** Restricting network diagnostics and socket analysis to loopback addresses (127.0.0.1 / ::1) ensures safety, compliance, and strict containment within the local machine stack during testing.

5. Report Questions & Answers

What is a Linux inode and why is it important?

An inode is a kernel data structure holding file metadata (permissions, timestamps, ownership, and disk block pointers) minus the file name. It is crucial in forensics because it persists independently of the file name, allowing analysts to track and recover files even after they have been renamed or deleted.

What is the difference between an absolute path and a relative path?

An absolute path specifies a location starting directly from the root directory (/), providing a fixed, unvarying reference regardless of the current working directory. A relative path specifies a location relative to the user's current active directory.

Why does Linux say “everything is a file”? Give two examples from /dev, /proc, /sys, or /etc.

Linux abstracts hardware devices, processes, system information, and configuration settings through the Virtual File System (VFS) as unified file streams.

- *Example 1:* /dev/sda1 represents a physical block storage device.
- *Example 2:* /etc/os-release contains operating system release configuration text.

What is the purpose of the PATH variable?

The PATH variable establishes an ordered list of directories that the command shell scans to locate executable binary files whenever a command is invoked, sparing the user from typing out full file paths.

Which grep command did you use to identify failed login evidence?

```
grep -n "failed" evidence/activity_log.txt
```

Why should a forensic trainee avoid scanning unauthorized external systems?

Scanning external networks or unauthorized systems without explicit authorization violates legal boundaries, breaches chain-of-custody protocols, and can trigger network disruptions or false alarms.

What is the difference between zip and tar.gz in this lab context?

zip packages files into a universally compatible compressed archive format. tar.gz creates a tape archive (tar) compressed with gzip (.gz), which natively preserves detailed Linux file permissions, ownership properties, and user attribute structures.

6. Challenges & Resolution

- *Challenge:* Certain automation utilities and network scanning packages were missing from the baseline installation.
- *Resolution:* Updated package repositories using `sudo apt update` and successfully staged the required tools using `sudo apt install -y tree nmap wget zip unzip tar net-tools terminator nano`.

7. Conclusion

This practical laboratory successfully demonstrated foundational command-line operations essential for digital forensics. Proficiency was gained in filesystem navigation, metadata evaluation via inodes, pattern-based searching, safe network diagnostics, script automation, and evidence archiving, building a strong base for upcoming digital forensics investigations.